

1. Introduction: Experimental Philosophy of Mind, Positive and Negative

Philosophers have used a variety of experimental methods to investigate topics squarely within the philosophy of mind. Indeed, in one of the earliest and most influential papers of contemporary experimental philosophy, Knobe (2003) investigated how people attribute the mental capacity of intentionality and uncovered a now famous asymmetry in our attributions of intentional states. Much experimental philosophy of mind has followed this early example, investigating folk theory of mind, in general, and when and why people attribute mental states of different kinds, in particular. In an important early paper, Knobe and Prinz (2008) investigated the attribution of mental states as such, and posited the existence of a folk psychological concept of phenomenal consciousness. They argued that such a posit was needed to explain their finding that people were less willing to attribute presumed phenomenal mental states, such as feeling sad, to group entities, such as corporations, than they were non-phenomenal mental states, such as thinking (whereas they were willing to attribute both kinds of states to other entities). In recent years, numerous other works of experimental philosophy have expanded and complicated our understanding of mental state attribution.

If one is inclined to accept that the identity conditions for at least some mental states are imposed by our intuitive folk psychological theories, then one will likely regard the results of the previous examples of experimental philosophy of mind as of instrumental value. Experimental philosophical projects of this kind purport to augment the philosopher of mind's toolbox with the tool of scientifically-informed, experimental studies of people's intuitions about mental states. Thus they can be seen as of a piece with more traditional, non-experimental, analytic philosophy of mind, since both have as their primary goal the investigation and further understanding of the metaphysics of minds and the epistemology of mental state attribution and since both typically

appeal to philosophical intuitions to advance their arguments.¹ Such projects, then, make positive contributions to analytic philosophy of mind.

However, a negative project also exists in experimental philosophy of mind. This project aims to undermine arguments that involve appeals to philosophical intuitions. Whereas positive experimentalists intend to augment the analytic practice of appealing to philosophical intuitions, negative experimentalists typically intend to show that some purportedly generally held philosophical intuition, to which some philosophical argument appeals, is not, in fact, generally held, or is not generally held in such a way as to support the argument it is intended to support. Because both non-experimentalists and positive experimentalists appeal to intuitions in making their arguments, either may end up as the target of a negative experimental philosopher of mind. Indeed, in one early paper, Sytsma and Machery (2009) argue that Knobe and Prinz's (2008) experiments on attributions of mental states to corporations suffer from a confound, and, therefore, that we need not posit a folk psychological concept of phenomenal consciousness to account for their findings. Other works of negative experimental philosophy challenge non-experimental arguments from the philosophy of mind. For example, many armchair philosophers of mind have argued that the existence and intractability of phenomenal consciousness is manifestly obvious. But Sytsma (2009) argues that "if phenomenal consciousness is phenomenologically obvious, then the concept of it should be shared by ordinary people" (967), and then he experimentally challenges the idea that philosophers and the folk are working with the same concept. These two papers demonstrate the two main ways in which negative experimentalists undermine arguments that involve appeals to philosophical intuitions within the

¹ Recently, a number of theorists have argued for the surprising thesis that analytic philosophers don't actually use intuitions as evidence, including Williamson (2007) and Cappelen (2012). However, a number of others have offered compelling challenges to this position, including Nado (2015), Climenhaga (2018).

philosophy of mind. Either 1) they attempt to show, as in Sytsma (2009), that there is some kind of *sampling bias* afoot—that while the target ‘intuition’ may be shared by some population (e.g., analytic philosophers, W.E.I.R.D. people), it is not shared by people at large, and is therefore not a pre-theoretic intuition we are committed to respect. Or 2) they attempt to show, as in Sytsma and Machery (2009), that there is a *confound in the design* of either a scientific or thought experiment—that, though the experiment might genuinely show how people behave, it doesn’t show that people have the intuition they are purported to have, because the results of the study are explicable without imputing that intuition to the folk. Negative experimental philosophy is negative from the perspective of analytic philosophy that accords evidential status to philosophical intuitions, but it also intends the positive effect of leaving analytic philosophy better able to address the questions it sets for itself by no longer relying on problematic intuitions.

In this chapter, I will focus, first and primarily, on the experimental philosophy of conscious state attribution. A mental state may be said to be conscious in a variety of ways, but experimental philosophers of mind have primarily been interested in phenomenal consciousness. As I explain below, experimental philosophers of mind have asked whether there are folk correlates to the philosopher’s concept of phenomenal consciousness and when and why we attribute phenomenally conscious mental states to some entities but not others. In Section 2, the main part of this chapter, I examine several different accounts of phenomenal state attribution that have emerged from the experimental philosophy literature. Throughout the discussion, the interplay of positive and negative projects will be central. In Section 3, I will briefly introduce several other topics of interest in the experimental philosophy of mind.

2. The Experimental Philosophy of Conscious State Attribution

Many theorists have distinguished between mental states that are, in some sense, conscious and other states that are not conscious, in that sense. For example, Freud claimed that repressed mental attitudes are not first-person accessible, and hence are unconscious, in a sense. At this point, many psychological studies have partially vindicated Freud and demonstrated that individuals often lack first-person awareness of various mental attitudes (beliefs, desires) that are, nonetheless, operative in their behavior.² The idea that people possess mental attitudes that are unconscious—in the sense of first-person inaccessible—is now widespread in western culture, to the point that we routinely attribute unconscious attitudes, as when we attribute implicit bias to someone.

However, many philosophers of mind have been interested in so-called phenomenal consciousness. According to Block (1995), “phenomenal consciousness is experience; what makes a state phenomenally conscious is that there is something “it is like” (Nagel, 1974) to be in that state” (228). Chalmers (1995) writes of the ‘subjective aspect’ of experience, claiming that sensory states, bodily sensations, emotions, and mental images are all united in that “...there is something it is like to be in them. All of them are states of experience” (201). Not only do philosophers of mind such as these countenance phenomenal consciousness, they further maintain that its existence is manifestly obvious to anyone who experiences phenomenally conscious states—that is, to all of us. For example, Chalmers (1995) goes on to claim that “experience is the most central and manifest aspect of our mental lives” (206). Jackson’s (1986) Knowledge Argument relies on the undefended premise that Mary will come to know about the

² Nisbett and Wilson (1977) covers numerous such studies at a very early date. More recently, arguments by analogy from research on commissurotomy (“split-brain”) subjects (see especially Gazzaniga, 1995, 2000) to the conclusion that we quite generally lack first-person awareness of our attitudinal states have been offered by Carruthers (2013, 39-45).

phenomenal experience of seeing red upon her release from her monochromatic internment.

Presumably, Jackson supposes that his readers will find this just obvious. Finally, these philosophers of mind are committed to the assumption that people attribute phenomenally conscious mental states to some but not all other entities. For example, Jackson (1986) goes on to claim that:

...after Mary sees her first ripe tomato, she will realize how impoverished her conception of the mental life of *others* *has been all along*...All along their experiences (or many of them, those got from tomatoes, the sky, . . .) had a feature conspicuous to them but until now hidden from her... (292-3).

And Nagel's (1974) "What is it Like to Be" Argument relies on our tacit acceptance that it *is* like something to be a bat. But, presumably, neither Jackson nor Nagel would maintain that we attribute phenomenally conscious states to a stone, a dead person, or a plant. Block's (1978) Nation of China Argument appeals to our intuition that the citizens of the nation of China, organized so as to instantiate the machine-table description of, say, the state of pain, would not thereby instantiate a pain state.

As these quotations suggest, some prominent analytic philosophers of mind suppose that there is a class of phenomenally conscious mental states that are obvious to those who experience them and are also attributed by those who experience them to some but not other entities. Inspired by these discussions of phenomenally conscious mental states, experimental philosophers of mind have investigated two main questions:

1. Do ordinary people possess a concept analogous to the concept of phenomenal consciousness employed by analytic philosophers of mind?
2. What features of an entity lead people to attribute the mental states that philosophers identify as phenomenally conscious mental states?

In the remainder of this chapter, I'll survey some projects that have attempted to answer one or both of these questions.

2.1 The Embodiment Hypothesis

An early inspiration for the experimental philosophy of conscious state attribution was provided by the psychologists, Gray, Gray, and Wegner (2007), who surveyed ordinary people's preferential attributions of different sorts of mental states to different sorts of entities. They had thousands of participants complete one survey each, consisting of 78 assessments as to which of two characters was more capable of a particular mental state. Participants selected the state they would assess, out of 18 possibilities (e.g., 'feeling pain', 'experiencing pride', 'exercising restraint', 'telling right from wrong'). Each participant assessed half of the 156 possible pairings of 13 different characters (e.g., an infant, a chimpanzee, a robot, a dead woman, and God). So, if a participant had selected the survey on the affective state of *pride*, they might be asked whether and to what degree a robot is more capable of experiencing pride than an infant, then asked whether a chimpanzee is more capable of experiencing pride than a person in a persistent vegetative state, and so on. Gray et al. subjected these results to a principle component factor analysis, and identified a two-factor model that accounted for 97% of the variance in people's assessments. One factor, which they called Experience, explained 88% of the variance and included capacities such as feeling fear and feeling hunger. Another factor, which they called Agency, explained 8% of the variance and included capacities such as self-control and moral reasoning.³ As shown in Figure 1, participants assessed some entities (a five-year old girl, the family dog) as more capable of Experiential states than some other entities (a robot, God). And

³ The Experience dimension was subsequently confirmed in replications by Takahashi, Ban, and Asada (2016) and Weisman, Dweck, and Markman (2017), though the Agency dimension has proved less robust. See Malle (2019) for further discussion.

they assessed some entities (God, an adult woman) as more capable of Agentic states than some other entities (a fetus, a dead woman).

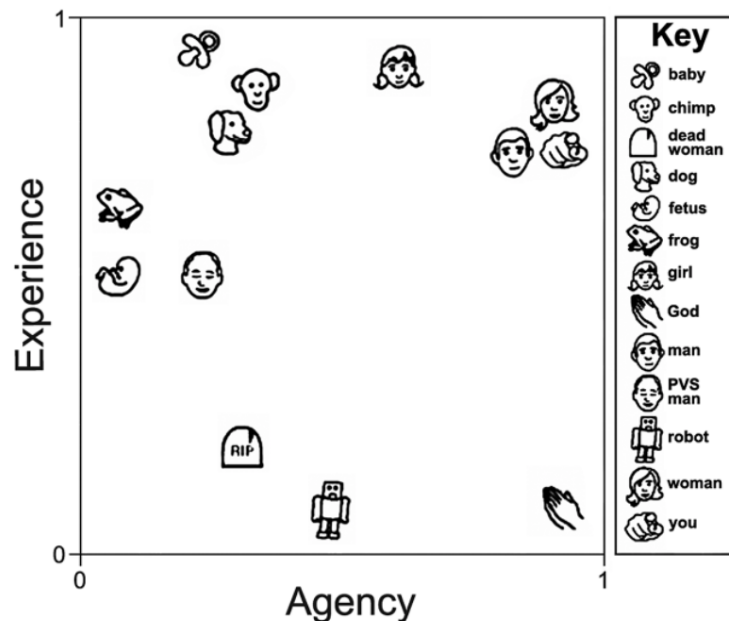


Figure 1. From Gray et al. 2007, showing “Adjusted character factor scores on the dimension of mind perception.”

One thing that stands out about Gray, Gray, and Wegner’s findings is that the entities assessed as least capable of experiential states—a dead woman, a robot, and God—are all alike in lacking a functioning biological body. But, despite lacking a biological body, God was still deemed highly capable of agentic states. Gray et al.’s experiential states are bodily sensations and emotions, so, they are prototypical phenomenally conscious mental states (and their agentic states are not). Gray et al.’s results, then, seem to suggest that considerations of bodily constitution weigh heavily in ordinary attributions of at least some phenomenally conscious mental states, but not in the attribution of other, non-phenomenally conscious mental states.

Building on these results, Knobe and Prinz (2008) used a series of experimental studies to argue that ordinary people possess an implicit concept of phenomenal consciousness and that “information about physical constitution plays a special role in those ascriptions that require phenomenal consciousness—a role it does not play in other kinds of mental state ascription” (71). Knobe and Prinz’s studies focused on people’s attributions of mental states to groups—to corporations, specifically. They pointed out that groups, being composed of individual agents, have a radically different constitution than a human being. Yet we seem willing to attribute, some mental states to group entities—we are perfectly willing to talk about what Apple thinks or what the Republican Party wants, for example. In one study Knobe and Prinz gave participants sentences that ascribed phenomenal states or (purportedly non-phenomenal) intentional states (such as believing or intending) to Acme Corp. They then asked participants to rate how natural each sentence sounded on a scale anchored at “sounds natural” and “sounds weird”. Group phenomenal state ascriptions sounded significantly weirder than group intentional state ascriptions to these participants. Knobe and Prinz inferred that these results “seem to indicate that people are unwilling to ascribe to group agents states that require phenomenal consciousness” (75). In another study, Knobe and Prinz showed that utterances such as:

- 1) a. Acme Corp. is feeling upset.
b. Acme Corp. is feeling regret.

Are judged to sound much weirder than utterances such as:

- 2) a. Acme Corp. is upset about the court’s recent ruling.
b. Acme Corp. regrets its recent decision.

They concluded from this result that, in addition to the folk psychological concepts of *regret* and *upset*, ordinary people also possess a concept of phenomenal consciousness, or *feeling*. And that, “...people’s reluctance to say that a group agent ‘feels upset’ stems not from the criteria

associated with their concept of upsetness, but rather from the criteria associated with their concept of phenomenal consciousness” (78).⁴

Reflecting on the results from Gray et al. (2007) and Knobe and Prinz (2008), we may ask ourselves what common feature a robot, God, and a corporation share that inhibits people’s attributions of phenomenally conscious mental states to them? According to Knobe and Prinz, this cannot be a feature related to how these entities function, since “a state of a corporation easily could have a functional role similar to one that people ordinarily associate with feeling depressed” (73). Intuitively, God or a robot could react to their environment in exactly the same way as an enraged person (though, see below for consideration of some other phenomenal states). At least *prima facie*, then, it seems that these three entities can occupy states functionally equivalent to the ordinary humans to whom we reflexively attribute phenomenal states. What God, a robot, and a corporation lack, however, and what a human possesses, is a unified, biological body. These findings, then, seem to support an Embodiment Thesis, which regards unified biological embodiment as a psychological cue to attributions of phenomenally conscious mental states. The Embodiment Hypothesis claims that ordinary people tacitly possess a concept of phenomenal consciousness roughly equivalent to the one philosophers of mind use and that they tend to apply it to entities that exhibit the feature of unified, biological embodiment.

However, Knobe and Prinz’s interpretation of their findings was quickly challenged by several other experimental philosophers. Heubner et al. (2009) presented versions of Knobe and

⁴ Knobe and Prinz use these and additional results to argue for a somewhat radical conception of human nature. They accept that information about the non-phenomenal, intentional mental states of other entities—information about their beliefs and desires—is useful for predicting and explaining behavior. But they maintain that we primarily exploit information about the phenomenal mental states of other entities to determine what our moral obligations to those entities are, and not primarily to predict and explain their behavior. If this is proposed as an adaptive explanation, it is tough to see what evolutionary or cultural advantage would have been bestowed in recognizing we have moral obligations to a fish (their example). In any case, since this further hypothesis is not particularly germane to our discussion, I set it aside.

Prinz's corporation probes to students at Hong Kong University, who were significantly more comfortable attributing phenomenal states to distributed, group entities than were their American counterparts. Embodiment tendencies, then, insofar as they exist, seem to be malleable and mutable, not universally held. Sytsma and Machery (2009) challenged the idea that people possess a concept of phenomenal consciousness that they express with the word *feeling*. They claimed that the sentence, "Acme Corp. is upset about the court's recent ruling," sounds more natural than the sentence, "Acme Corp. is feeling upset," simply because the mental state attribution is contextualized with a prepositional phrase in the former but not the latter. They were able to reverse the effect Knobe and Prinz found, getting participants to judge that the sentence, "Acme Corporation is feeling upset about the court's recent ruling," sounds more natural than the sentence, "Acme Corporation is upset." Similarly, Arico (2010) found no difference between people's naturalness assessments of "McDonalds is upset" and "McDonalds is feeling upset" when each was contextualized with the addition of the prepositional phrase "about the court's recent ruling". Finally, Phelan et al. (2013) found that participants were equally likely to choose 'they' (rather than 'it') to paraphrase a group name in group phenomenal state attributions and group intentional (non-phenomenal) state attributions. They were significantly more likely to choose 'it' to paraphrase a group name in non-mental state attributions. This suggests that people understand *all* group mental state attributions—phenomenally conscious or not—distributively, as attributions of states to the constituents of groups. If correct, Knobe and Prinz's results tell us very little about how people think about the mental states of groups, as opposed to how they think about the people who constitute groups.

Concurrently, with these negative projects, positive projects within experimental philosophy of mind seemed to demonstrate that people are, in fact, willing to attribute

phenomenally conscious states to entities that lack a biological body. Sytsma and Machery (2010) found that people were willing to attribute states that philosophers would regard as involving phenomenal consciousness—seeing red, smelling isoamyl acetate—to a simple robot. However, their participants seemed unwilling to attribute the states of feeling pain, smelling vomit, or smelling bananas to the robot. They argued from these results to the conclusion that ordinary people assess mental states according to whether or not they have a positive or negative valence. Buckwalter and Phelan (2014) showed that people were willing to attribute happiness and sadness to spirits and ghosts. While functional information greatly affected people's emotional state attributions, they found no effect for embodiment when responses for disembodied entities were compared to responses for a normally constituted person. Phelan and Buckwalter (2012) argued from these and other results that people attribute mental states (conscious or not) primarily through reference to functional information about the target entities. These projects will be the focus of sections 2.2 and 2.3 respectively.

Despite these challenges to the Embodiment Hypothesis, the role that an entity's body plays in our attributions of emotions, perceptual states, bodily sensations, and other phenomenally conscious states has remained a topic of interest. For example, Forstmann and Burgmer (forthcoming), have recently purported to provide evidence that people conceptualize consciousness as a spatio-temporally located process in the human brain. Though, it is doubtful that their method of asking people to color the part(s) of a two-dimensional picture of a brain that they think “contribute” to consciousness delivers much evidence as to where people think consciousness is located, since something can clearly contribute to a process without being collocated with it. Meanwhile, Diaz (2021) has presented experimental evidence that “people report anger, sadness, and fear in the absence of bodily feelings” (24). If correct, this suggests

that bodily responses are not essential to ordinary people's conception of emotions and, thereby undermines one source of support for a sort of embodiment view.

2.2 The Valence View and the Naïve View

The Embodiment Hypothesis answered the first of experimental philosophy of mind's key questions affirmatively. People tacitly employ a concept of phenomenal consciousness analogous to that employed by analytic philosophers of mind, it claimed. Further, it maintained in answer to the second question that when we decide to predicate phenomenally conscious mental states to entities, we do not merely consider the functional capacities of those entities. We also consider physical properties that are irrelevant to functional capacities. The Valence View and the Naïve View, however, are alike in giving a negative answer to the first of experimental philosophy of mind's key questions. Both maintain that ordinary people do not employ a concept of phenomenal consciousness analogous to that employed by analytic philosophers of mind. Instead, they contend that the diverse set of sensory states, bodily sensations, emotions, and mental images that analytic philosophers have identified as phenomenally conscious are conceptualized in a variety of different ways by ordinary people, no one of which approximates the analytic philosopher's concept of a phenomenally conscious mental state. Both views maintain that the analytic philosopher's commitment to the manifest obviousness of phenomenal consciousness was, in fact, the product of biased post-theoretic reflection on what is pre-theoretically obvious. However, none of this means that the Valence View or the Naïve View are off the hook when it comes to the second question. In fact, they have complicated the task, since either view would seem to entail that we need a variety of different accounts for how people apply each of the motley set of mental states philosophers mistakenly unify under the heading 'phenomenally conscious states'.

According to the philosophically prominent conception of phenomenal consciousness described above, we must accept the existence of phenomenal consciousness because its existence is obvious to anyone who experiences it and we all continually experience it. Proponents of this prominent conception have, further, identified both bodily sensations (e.g., the experience of physical pain) and sensory states (e.g., the state of seeing something red) as prototypical instances of phenomenally conscious mental states. According to this view, then, it seems that anyone who occupies the mental state of physical pain should experience that state as phenomenally conscious—that is, as a state that it is like something to occupy. And the same follows for anyone who occupies the state of seeing something red. Presumably, given the obviousness of phenomenal consciousness, after a little experience of bodily sensations and sensory states like these, most everyone would associate the two sorts of mental states together as being phenomenally conscious—that is, as being like something to occupy. Sytsma and Machery (2010) reason that, if all of this is the case, then ordinary people should tend to treat attributions of these paradigmatic examples of phenomenally conscious mental states similarly and similarly to how people with philosophical training treat them.

With this argument in mind, Sytsma and Machery designed a set of vignettes to test whether ordinary people attribute bodily sensations and sensory states in the same way as philosophers. In their first study, Sytsma and Machery split their sizable sample into two groups: “Philosophers”—including undergraduate majors and participants with some graduate training in philosophy—and “Non-philosophers”. Participants in each group were presented with one of four vignettes. Two vignettes described a normal human (Timmy), the other two described a simple, non-humanoid robot (Jimmy). In every vignette, Timmy or Jimmy was instructed to place a red box in front of a door as part of a psychological experiment. Timmy or Jimmy

successfully completed the task in two of the vignettes. In the other two, the box gave Timmy or Jimmy “a strong electric shock” and, “He let go of the box and moved away from it” (##). For each vignette, the participant was asked whether Timmy or Jimmy ‘saw red’ or ‘felt pain when he was shocked’, and answered on a 7-point scale anchored at 1 with “clearly no,” at 4 with “not sure,” and at 7 with “clearly yes.” As the results in figure 2 show, the responses of the philosophers diverged from those of the non-philosophers when it came to the purported phenomenal states of seeing red.

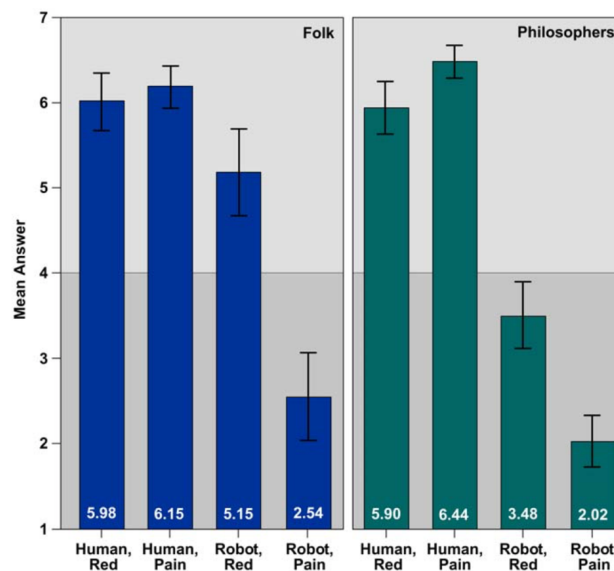


Figure 2. Results of Sytsma and Machery (2010) Study 1.

As Sytsma and Machery report, philosophers’ mean attributions of the mental states of feeling pain and seeing red were significantly above the midpoint for Timmy, the normal human, and significantly below the midpoint for Jimmy, the simple robot. For non-philosophers, mean attributions of both mental states were significantly above the midpoint for Timmy, as was the mean attribution of seeing red to Jimmy. Only attributions of feeling pain to Jimmy were below the midpoint for non-philosophers.

On the basis of their first study, Sytsma and Machery tentatively conclude that non-philosophers lack the philosopher's concept of phenomenal consciousness. They then strengthen this tentative conclusion with a series of experiments that serve to develop their positive view of how people think about sensory states and bodily sensations. They ultimately conclude that philosophers and ordinary people agree that simple robots are not capable of what we may describe as subjectively rich experiences. It's just that philosophers and ordinary people part ways on what counts as 'subjectively rich'. Philosophers identify the category of phenomenally conscious mental states with subjectively rich experiences. Ordinary people, on the other hand, sort experiential states according to their valence and identify positively or negatively valenced mental states with subjectively rich experiences.⁵

This Valence View explains Sytsma and Machery's findings. Because feeling pain is negatively valenced, ordinary people, like the philosophers operating with their concept of phenomenal consciousness, regard this as a subjectively rich experience to be denied a simple robot. But ordinary people are willing to attribute the state of seeing red to the simple robot because they do not regard it as a subjectively rich (that is, valenced) experience. Philosophers, though, are unwilling to attribute seeing red to the robot, because they see it as a subjectively rich (that is, phenomenally conscious) experience.⁶ Bolstered by their Valence View of how ordinary people think of sensory states and bodily sensations, Sytsma and Machery (2010) conclude that, "in clear contrast to philosophers, the folk do not seem to believe that there is something common to all these mental states—namely that they are phenomenal" (319).

⁵ Sytsma and Machery (2010) helpfully clarify their use of the term 'valence' in a footnote: "mental states have a valence if and only if they have a hedonic value for the subject. That is, mental states have a valence if and only if they are pleasurable (they then have a positive valence) or disagreeable (they then have a negative valence). Not all mental states have a valence, and valenced states are more or less pleasurable or disagreeable" (350).

⁶ A similar explanation can be applied to the results Sytsma and Machery find in their other studies for the mental states of smelling vomit, bananas, and isoamyl acetate.

Shortly after the publication of Sytsma and Machery (2010), several studies appeared which presented challenges for the Valence View. For example, Buckwalter and Phelan (2013), designed a pair of studies based on the simple robot vignettes. In their first study, they asked participants whether a robot smelled something, either isoamyl acetate, bananas, or vomit. These sensory states were chosen because, in one of their studies, Sytsma and Machery had found that participants were more inclined to say the robot smelled isoamyl acetate (presumably unknown to these participants, and, so, unvalenced) than either vomit or bananas. However, Buckwalter and Phelan varied the functional complexity of the robot and the functional role the robot was designed to perform across conditions. They also had participants rank the valence of the three sensory objects and found that participants attributed little positive or negative valence to isoamyl acetate, a slight positive valence to bananas, and a strong negative valence to vomit. They then compared the degree to which participants rated the valence for each smell with the degree to which they attribute the subjective experience of smelling the object to the robot. Contra the Valence View, no relationship was found between the intensity of positive or negative valence ratings and smell attributions. However, there was a strong relationship between the functional role a robot was designed to perform and people's attributions of sensory states. When the robot was designed to make smoothies, people were more likely to say that he smelled the chemical and the banana but not the vomit. Conversely, when his functional role was to clean up biomedical waste, mean scores were higher for vomit than for the chemical (and to a lesser extent the banana). In their first study, Buckwalter and Phelan did not find an effect for complexity of the robot. However, in a second study, participants' assessments of whether or not a robot could feel guilt were influenced both by its complexity and its specified functional role. But, again, in this study, participants valence assessments for the emotional state of guilt were

not predictive of their guilt attributions. Finding no evidence of a valence effect, Buckwalter and Phelan explained Sytsma and Machery's results by referencing participant's assumptions about function, rather than valence. Specifically, they argue that participants likely assumed that a simple robot in a psychological study that was directed to move a red box was designed to satisfy the functional role of detecting colors and with the mechanical complexity to do so. Such a robot was probably not designed to feel pain when it is damaged, especially since the shock it sustains is likely to be interpreted as the result of an accident, rather than a planned intervention by its designers.⁷

To his credit, after his own studies failed to present evidence in favor of the Valence View, Sytsma (2012) abandoned it, arguing from previous research and original experiments to the conclusion that ordinary people tend to "hold a naïve view of both colors and pains" (190). According to this Naïve View, ordinary people take "colors to be mind-independent qualities of worldly objects" and pains "to be mind-independent qualities of certain body parts" (188). The Naïve View offers an explanation of Sytsma and Machery's findings. Ordinary people thought the simple robot was incapable of feeling pain but not seeing red because they thought that it could see the mind-independent color inherent in the box in front of it, whereas they also thought it lacked the appropriate, fleshy body, in which a pain state might occur.⁸ Sytsma (2012) offers three additional studies supporting the Naïve View. The first reveals that participants are more

⁷ See also Fiala, Arico, and Nichols (2014), who provide evidence that few people describe the robot in Sytsma and Machery's (2010) vignettes as 'seeing green' when given options to describe it, instead, as 'detecting green' or 'identifying the green box' (whereas most people describe the person as seeing green). As they explain, "These results suggest that the overly narrow choice format might artificially inflate the attributions of perceptual experience to robots in Sytsma and Machery's (2010) studies" (39).

⁸ While the Naïve View doesn't explain Buckwalter and Phelan's (2013) findings, those findings are at least compatible with the Naïve View. Even if people think colors exist out in the world to be seen, as the Naïve View contends, they presumably think an entity needs the right sort of functional machinery (e.g., eyes) in order to see colors. Participants' responses that a robot designed to make smoothies couldn't smell vomit, then, might be due to the fact that they don't think such a robot would have the right kind of functional machinery to experience the smell of vomit, even though they think the vomit smell is actually out there in the world to be experienced.

willing to attribute pain to someone who burns their bioengineered, fleshy prosthetic hand, than to someone who burns their robotic prosthetic hand, supporting the idea that people tend to view pain as a mind-independent quality of certain kinds of body parts. In the second, he finds that participants are just as willing to attribute the experience of seeing red to someone with robotic eyes as to someone with bio-engineered eyes, suggesting that bodily constitution doesn't matter for seeing red. In the third study, he shows that the results for pain generalize to robot agents, by showing that participants are more willing to attribute pain states to a robot agent with fleshy, bioengineered hand than to a robot agent with hard, metallic hands.

While the Naïve View is similar to the Embodiment Hypothesis in advocating that considerations of physical makeup play an important role in ordinary attributions of some prototypical phenomenal mental states, it differs from the Embodiment Hypothesis in claiming that this is true only for pain states and other states that ordinary people regard as adhering in particular types of bodies. The Naïve View, like the Valence View, and contra the Embodiment Hypothesis, also maintains that ordinary people lack a concept of phenomenal consciousness, for at least two reasons. First, as Sytsma (2012) concludes:

While philosophers tend to treat the qualities at issue for ascriptions of seeing red and feeling pain as mind-dependent qualia...lay people understand [these] two prototypical examples of (supposed) phenomenally conscious mental states in a way that is fundamentally different from the standard philosophical account (195).

Second, though not noted by Sytsma (2012), while a Naïve View may be plausible for the supposed phenomenal states of seeing red or feeling pain, it is implausible to suppose that other states that philosophers identify as prototypically phenomenal are thought by ordinary people to be mind-independent qualities of worldly objects. In what worldly object or part of the body would sadness reside, for example? What about a hallucination of a pink elephant? Thus,

ordinary people do not think of all the states that philosophers group under the heading phenomenally conscious mental states in the same way.

In other papers, Sytsma and colleagues have deepened the case for the Naïve View of colors and pains. Sytsma (2010) included numerous findings that could be seen as consistent with the Naïve View. In one study, participants tended to answer in a way that suggested they think of the redness they experience when they see a ripe tomato as not in their minds, but out there, on the tomato, an objective property that is sure to be experienced in the same way by any normal-sighted individual. In another set of studies, participants tended to respond that two agents who share a body part (e.g., conjoined twins) experience the same pain when they injure their shared part, challenging the analytic philosopher of mind's conception of pain as a private mental state locked away in each individual's mind. In another set of studies, Sytsma (2010) found that participants were likely to judge there was an unfelt pain, when presented with vignettes describing seriously injured patients who reported they hadn't been aware of pain while watching an engrossing movie or reading a compelling book, suggesting that ordinary participants don't think of pain as a necessarily conscious state. Reuter & Sytsma (2020) deepened this last result through 17 additional studies, which demonstrated that ordinary participants are happy to say that people experience unfelt pains. In another vein of research, Reuter, Phillips, and Sytsma (2014) offered a series of studies suggesting that ordinary people think it's possible to hallucinate a pain—i.e., to have an experience as of a pain when no pain is, in fact, present. This accords with a perceptual model of pains, since perceptions are subject to an appearance-reality distinction, and it runs counter to the views of analytic philosophers of mind, such as Hillary Putnam (1963), who claims that “any situation that a person cannot discriminate from a situation in which he himself has a pain *counts* as a situation in which he has

a pain” (218). An appearance-reality distinction for pain is further supported by an argument Reuter (2011) and Sytsma and Reuter (2017) make from their findings that ordinary English and German speakers preferentially use the phrase ‘having pain’ (or its German analogue) to express mild pains and ‘feeling pain’ (or one of its three German analogues) to express large pains. If we can take these attributions literally (as Sytsma and Reuter argue we can), then ordinary people seem to acknowledge pains you have but don’t feel. In yet another vein of research, Reuter et al. (2019) present a challenge to the ‘finger in mouth’ argument against the Naïve View of Pain. According to that argument, if a pain is actually located in my finger (as the Naïve view would claim) and I put that finger in my mouth, then it should follow that I have a pain in my mouth. But, it seems incorrect to say that I have a pain in my mouth in that situation. Reuter et al. contend that, even though it follows that one does have a pain in one’s mouth in this situation, we would be hesitant to say that, because having a pain in a body part—like having tissue damage in a body part—tends to imply something (a pain) adhering in the substance of the body part, but the premises of the argument do not suggest that there is pain adhering in the mouth’s substance. They provide evidence that, if you cancel these implications, people are happy to say there is a pain in the person’s mouth. Finally, Kim et al. (2016) began to make the case that the naïve conception of pain is a cultural universal, by presenting data from two studies suggesting that Americans and South Koreans “overwhelmingly conceive of pains as bodily states” (163).

At this point, proponents of the Naïve View have compiled a great deal of evidence for the claim that ordinary people sometimes think of pains as mind-independent qualities of certain body parts and a bit of evidence for the claim that they sometimes think of colors as mind-independent qualities of worldly objects. But what exactly is the scope of these claims? Is it simply that ordinary people *sometimes* think of pains and colors in a mind-independent way? Or

is it the much stronger claim that ordinary people *only* think of pains and colors in this way? Or is it some middle view that is being suggested? Unfortunately, the above papers are not very clear on this point. Sytsma and Reuter (2017) is typical. In the conclusion, the authors claim that, “the accumulated research suggests that people tend to conceive of pains not as mental states, but as bodily states” (23), which seems to suggest a folk *tendency* towards the Naïve View, but to also leave open the possibility that people sometimes think of pains in other ways. In fact, this conclusion seems to be well supported by the data, since, in each study on pain attributions a (sometimes sizeable) minority of participants seemed to endorse a mentalistic view of pain. However, Sytsma and Reuter (2017) also refer to the Naïve View as “*the* commonsense conception of pains” (23) [italics added]. And if proponents of the Naïve View want to maintain, as Sytsma (2012) did, that ordinary people “lack the philosophical concept of phenomenal consciousness” (183), then they will have to adopt the additional assumption that the common-sense concept of pain they have identified operates to the exclusion of other, mentalistic conceptions of pain. But, Sytsma and colleagues have not defended this additional assumption. And recent work by Borg et al. (2019) and Saloman et al. (2021) suggests considerable conceptual diversity in the ways in which ordinary people can and do think of pain.

2.3 Functional Considerations and the Agency Model

Though radically different in a variety of ways, the previous accounts of phenomenal state attribution are alike in offering definitive (though distinct) responses to the question of whether ordinary people possess a concept of phenomenal consciousness. They are also alike in applying only to attributions of states that philosophers would consider phenomenally conscious (or to subsets of those states), not to attributions of mental states in general. The views we will consider in this section differ from the previous ones on both counts. These views claim that we are not

yet in a position to say whether people possess a concept of phenomenal consciousness. They also claim that our attributions of states that philosophers would consider phenomenally conscious are guided by the same sorts of considerations that guide our attributions of mental states in general.

It is worth noting that, before the analytic philosopher's conception of phenomenal consciousness came to prominence, many philosophers tended to emphasize quite different aspects of our ordinary understanding of experiential states such as pain. These philosophers did not emphasize the, purportedly obvious, qualitative aspects of experiential states, instead they suggested that ordinary conceptions of experiential states (like other mental states) centered on the functional roles those states played. So, for example, Lewis (1966) contends that, "...the definitive characteristic of any experience as such is its causal role. The definitive causal role of an experience is expressible by a finite set of conditions that specify its typical causes and its typical effects under various circumstances" (19-20). And he goes on to assert that this is, "...an account of the parlance common to all who believe that experiences are something or other real and that experiences are efficacious outside their own realm" (20). Armstrong (1968) suggests, "that what we mean when we talk about the mind, or about particular mental processes, is nothing but the effect within a man of certain stimuli, and the cause within a man of certain responses" (79). Braddon-Mitchell and Jackson (1996) offer a succinct, yet detailed, statement of this sort of view of ordinary mental state concepts:

We distinguish the roles that matter for having a mind, and matter for being in one or another mental state, by drawing on what is common knowledge about mental states. We extract the crucial functional roles from the huge collection of what is pretty much common knowledge about pains, itches, beliefs, desires, intentions, and so on and so forth, focusing on what is most central to our conception of what a pain is, what a belief is, what it is to desire beer and all the rest. And we can group the common knowledge into the three clauses distinctive of the functionalist approach. The input clauses will contain sentences like 'Bodily

damage causes pain' and 'Chairs in front of people in daylight cause perceptions as of chairs'; the output clauses will contain sentences like 'Pain causes bodily movement that relieves the pain and minimizes damage' and 'Desire for beer causes behavior that leads to beer consumption'; the internal clauses will contain sentences like 'Perception as of beer in front of one typically causes belief in beer in front of one' and 'Belief that if p then q typically causes belief that q on learning p ' (46).

Philosophers who have introduced this conception of our ordinary understanding of mental states have sometimes used it to argue that the entities that occupy the functional roles specified by our common-sense understandings of mental states are brain states, and thereby defended an identity between minds and brains (as do Armstrong, 1968, and Lewis, 1966). As Fiala et al. (2014) point out, "it is doubtful whether ordinary people are committed (even implicitly) to a theory like this" (45, fn. 3). However, those who suggest a functionalist answer to the question of what features lead people to attribute phenomenally conscious mental states do not attribute this metaphysical theory to ordinary people. Instead, they use empirical methods to argue for two claims:

1. We have good evidence that people exploit information about the functional roles an entity can and does occupy in deciding what mental states (including phenomenally conscious states) it is (or is capable of) occupying, and;
2. We have no good evidence that people exploit information that does not contribute to information about the functional roles an entity can and does occupy in deciding what mental states it is (or is capable of) occupying.⁹

If these two claims are correct, they contend, it would be unsurprising if it turned out that people's attributions of mental states—phenomenal states included—were simply mediated by their beliefs about an entity's functional capacities and the functional roles it currently occupies.

⁹ Information that contributes to information about functional roles is here understood to include information about functional roles and any other information that implies information about functional roles. For example, information about the purpose for which some device was designed may imply information about the functional roles it might occupy.

Many of the results considered previously could be interpreted so as to provide indirect evidence for the first of the functionalist's two claims. In replying to Knobe and Prinz (2008), Sytsma and Machery (2009) and Arico (2010) both found that the addition of a prepositional phrase further contextualizing a mental state attribution led participants to judge that attribution more natural. One explanation of this finding is that, for example, adding "about the court's recent ruling" to the attribution "McDonald's is feeling upset" provides participants with an input condition—negative court rulings cause feelings of upsettedness—and thereby leaves them more willing to attribute their functional concept of 'feeling upset'. As alluded to previously, we might offer a similar, alternative explanation of Sytsma and Machery's (2010) robot vs. human findings. Whether or not an entity can satisfy a specific, functional input condition depends in part on considerations of that entity's functional capacities. As Braddon-Mitchell and Jackson suggest in the excerpt above, chairs in front of people in daylight are taken to cause perceptions as of chairs. But, of course, chairs in a forest devoid of anything but trees are not taken to cause perceptions as of chairs, even when the sun is shining. Obviously, this is because trees are not generally regarded as the right sorts of entities to satisfy the functional input conditions for visual states. They are supposed to lack the sensory apparatus essential for visual experiences. Sytsma and Machery may have inadvertently manipulated people's assumptions about the sort of sensory apparatus their simple robot was equipped with by putting it in the context of what appeared to be a psychological study about vision. People may have supposed such a robot possessed the sensory apparatus essential to seeing red, and thus regarded it as capable of satisfying the functional input conditions for seeing red. However, the vignettes give us no reason to suppose the robot was designed with whatever apparatus is necessary for feeling pain. Buckwalter and Phelan (2013) subsequently demonstrated the plausibility of such an explanation, by showing

that manipulating the role a simple robot was designed to perform significantly affects people's attributions of sensory and emotional states to that robot. Presumably, this is because it affects their assumptions about the functional capacities such a robot will have. It tells the participants something about whether the robot is more like a person presented with a chair or a tree in a similar situation.

Several studies have also provided direct evidence for the first claim. For example, Huebner (2010), had participants assess their agreement with various mental state attributions to four sorts of entities: A normal human with a Central Nervous System (CNS); A human with a CPU in place of a CNS, A robot with a CNS in place of a CPU, and a robot with a CPU. In a reasonable attempt to standardize the functional capacities of each of these entities (and so to isolate the role of constitution) Huebner described each as behaving "in every respect like a person on all psychological tests" (138). He found that while people were somewhat inconsistent in their attributions of some paradigmatic phenomenal states to either the robot or the cyborgs, they strongly agreed with attributions of non-phenomenal, intentional states to all four entities. This suggests that people exploit functional information in attributing intentional mental states, at least. Additional evidence that people exploit information about functional roles in deciding what mental states an agent is capable of occupying is provided by Buckwalter and Phelan (2014). They investigated people's attributions of the emotional states of happiness and sadness to embodied and disembodied agents across five studies. While they found no effect for whether or not the agent was embodied in any of these studies, each study revealed a strong effect for how the entity was functionally described. In their third study, for example, participants were presented with a vignette describing either a hermit or an entirely disembodied eternal forest spirit living in a forest. The agent was then described as responding to encroaching loggers by

breaking their equipment. Finally, the loggers were either turned away by the retaliation or else the agent (the hermit or spirit) was driven out of the forest by the loggers. As figure 3 shows, participants strongly agreed with attributions of feelings of sadness to the embodied hermit *or* the disembodied spirit in the sad situation, in which they were driven from their forest home. They did not attribute sadness in the happy situation, in which the functional role the agent occupies is more appropriate to sadness. Conceptually equivalent results were found for happiness.

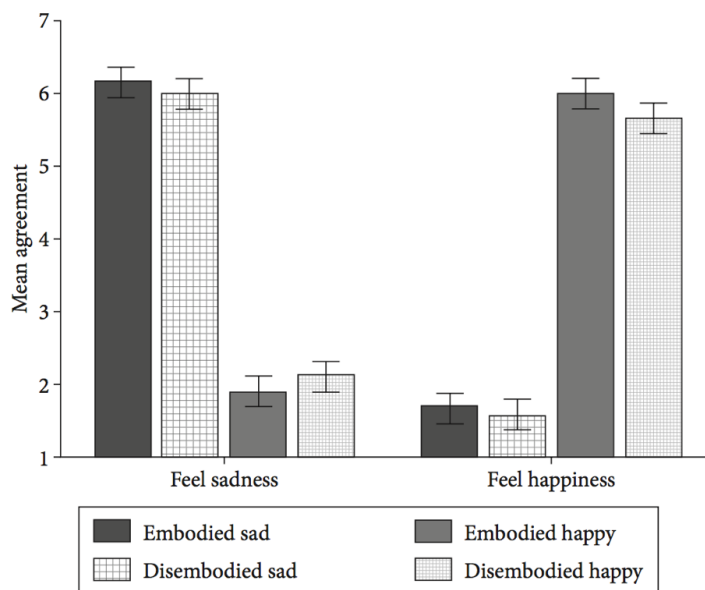


Figure 3. Assessments of attributions of feelings of sadness/happiness for an embodied/disembodied sophisticated entity in a happy or sad situation from Study #3, Buckwalter and Phelan (2014).

These results suggest that if an unknown agent is described as being more like a person than a tree and comes to occupy an appropriate input condition for a particular emotional state, then

ordinary people will find it appropriate to describe that entity as occupying that emotional state, regardless of how that entity is physically constituted. In other words, people exploit information about the functional roles an entity can and does occupy in deciding what phenomenally conscious mental states it is capable of occupying. And they seem to prioritize this information over other sorts of information about the entity (e.g., about the entity's physical constitution) in attributing emotional states.

It seems undeniable—and, indeed, I doubt anyone would deny—that people exploit information about the functional roles an entity can and does occupy in deciding what mental states it is (or is capable of) occupying. However, the functionalist's second claim asserts something additional, specifically, that there isn't good evidence that people exploit information that does not contribute to information about functional roles in deciding what mental states an entity is capable of occupying. But how can the functionalist defend this claim in light of the studies above that seem to suggest that people exploit non-functional information about entities in deciding what mental states to attribute to them?

As alluded to previously, functionalists have responded to some of these studies by presenting empirical evidence of their own to support a functionalist interpretation of results initially used to promote an anti-functionalist account of mental state attributions. For example, in the course of responding to Knobe and Prinz's (2008) results that people found attributions of non-phenomenal, propositional attitudes to groups less weird than they found attributions of phenomenal mental states, Phelan et al. (2013) presented their own results that participants were significantly more likely to choose 'they' paraphrases for group names in sentences that attributed a phenomenal *or* non-phenomenal mental state to a group, compared with sentences that attributed a non-mental state. On the basis of these two sets of results, Phelan et al. then

argued that ascriptions of mental states to groups are typically understood distributively, as attributions of states to the members that make up the relevant group *qua* members of that group. They contended that mental states are thought to be appropriately attributed to group members *qua* members on functional grounds, insofar as the relevant input, output, and internal clauses associated with the relevant mental state are saliently associated with the role of being a member of the relevant group. If correct, Knobe and Prinz's (2008) findings are explicable functionally, along the following lines: Participants found it appropriate to say that 'Acme Corp. believes that its profit margin will soon increase' or 'intends to release a new product this January' because they understood these as attributions to the employees that make up Acme Corp. in their role as Acme Corp. employees and they associated beliefs about profit margins and intentions to release products with the role of being a corporate employee. They also understood 'Acme Corp. is getting depressed' or 'is feeling excruciating pain' as attributions to the employees that make up Acme Corp. in their role as Acme Corp. employees, but they did not associate feelings of depression or excruciating pain with the role of being a corporate employee, and so they judged these attributions as inappropriate. If this functionalist explanation is correct, then we might expect participants to find attributions of phenomenal states to groups such as ISIS more appropriate, since phenomenal states are more saliently associated with being a member of such a group. Similarly, we discussed experiments supporting a functionalist interpretation of Sytsma and Machery (2010) in Section 2.2.

As the previous discussion indicates, there are too many studies purporting to show that people exploit non-functional information in attributing mental states to make an experimental response practical in every instance. Theorists have thus, on occasion, argued for the plausibility of a functionalist interpretation of some results or other in the absence of experimental support.

Phelan and Buckwalter (2012) contains a number of such arguments. They argue, for example, that Gray et al.'s (2007) finding that God was deemed less susceptible to experiences than a host of other entities was the result of people's functional beliefs about the kinds of inputs and outputs to which an all-powerful God is subject, rather than their beliefs about God's physical constitution. After all, Gray et al.'s study describes God as "the creator of the universe and the ultimate source of knowledge, power, and love". But it is unsurprising for functional reasons that such a God was deemed less susceptible than a host of mere mortals to hunger, fear, pain, pleasure, and rage.¹⁰ Think of the many input and internal clauses for fear to which a mere mortal is susceptible that make no sense to apply to an all-powerful, eternal God—e.g., considerations of one's own mortality cause fear, but considerations of one's own mortality are off the table for God. Similar functionalist explanations are possible for other of Gray et al.'s experiential states, and, indeed, for other of the studies discussed previously.

To these specific arguments for a functionalist interpretation of previous results, we can add a general consideration: It is unclear how experimental manipulations to an entity's physical constitution impact people's understanding of the sorts of functional clauses to which that entity is subject. Clearly, *some* manipulations to physical constitution influence what functional clauses we think an entity capable of satisfying. Consider the input clause for visual sensations discussed previously, 'Chairs in front of people in daylight cause perceptions as of chairs.' It seems likely that we accept such an input clause for people *in general*, but, intuitively, we wouldn't accept such a clause for a person without eyes. This is because we suppose that eyes are required for certain functional capacities related to visual states. So, those without eyes are incapable of satisfying this input clause and others like it, and, if satisfying such input clauses is requisite for

¹⁰ Malle (2019) points out that these were the highest loading experiential traits in Gray et al. (2007).

having visual experiences, then those without eyes are incapable of having visual experiences on functional grounds. However, little is known about what functional capacities ordinary people think are requisite for different sorts of functional roles, or about how they think specific aspects of physical constitution bear on functional capacities. Do people think fleshy hands are required to occupy the functional roles associated with haptic sensations? Do they think stomachs are required to occupy the functional roles associated with feeling hungry? We don't know the answer to questions such as these, so we don't know how manipulations of an entity's physical constitution influence ordinary people's understanding of that entity's functional capacities. And experimental philosophers have not typically included probes to assess people's inferences about functional capacities. As a result, we don't know if purportedly anti-functional results are in fact mediated by inferences about functional roles.

If the foregoing arguments are correct, then we seem to lack good evidence that people exploit information other than that which contributes directly or indirectly to information about functional roles in assessing an entity's mental states. On the other hand, we seem to have positive evidence that considerations of functional role influence mental state attribution. At our present stage of inquiry, then, it would be unsurprising if it turned out that people's attributions of mental states—phenomenally conscious or not—were simply mediated by their beliefs about an entity's functional capacities and the functional roles it currently occupies.

An intriguing twist to these functional considerations is offered by Arico et al. (2011) and Fiala et al. (2014), in the form of their Agency Model of mental state attribution. In Arico et al. (2011), participants were subjected to a reaction time study in which they agreed or disagreed with various state attributions—including phenomenally conscious state attributions ('feels anger', 'feels happy', 'feels pain', 'feels pride')—to various entities (Mammals, Birds, Insects,

Plants, Artifacts, Vehicles, Inanimate Natural Objects such as mountains, and Moving Natural Objects such as clouds). They found that significantly more people responded affirmatively to phenomenal state attributions to insects than to plants, vehicles, or natural moving objects. They also found that it took participants longer to deny phenomenal states to insects than to vehicles or natural moving objects. On reflection, it may seem somewhat surprising that people were willing to attribute states such as feeling anger or pride to insects at all. However, Arico et al. (2011) argue on the basis of these results for a dual process model of mental state attribution. And Fiala et al. (2014) support this dual process model through a wider survey of the literature on mental state attribution. According to these researchers' Agency Model, there is a high road and a low road to mental state attribution. When asked for their considered judgment about whether an entity has mental states (as in the previous vignette studies), people depart on the high road and end up producing a reasoned assessment based, in large part, on functional considerations. But, if forced to make a speeded judgment (as in Arico et al., 2011) or a judgment under cognitive load, people go down the low road and manifest a more fundamental tendency to attribute a whole host of mental states—including fairly complex emotional states—to any entity that triggers the 'agent' concept, which is triggered by any entity that possesses one or more simple features, including eyes, particular motion trajectories, or contingent interaction. These researchers, then, break new ground in the experimental philosophy of conscious state attributions without really challenging any of the previously considered views.

The Functionalist and Agentic views are alike in claiming that our attributions of phenomenal states to other entities are guided by the same sorts of considerations that guide other of our mental state attributions. The Functionalist emphasizes only functional considerations, whereas proponents of the agentic view suggests that simpler considerations

come into play in more automatic, unreasoned responses. These views are also alike in not taking a stand on the question of whether ordinary people draw a distinction between phenomenal and non-phenomenal mental states. They only contend that such a distinction is not supported by considerations of how people attribute mental states.

2.4 Further Reading on the Experimental Philosophy of Conscious State Attribution

Huebner (2015) raises a compelling challenge for the previous instances of—and, indeed, most extant—experimental philosophy. He argues that

3. Other Work in the Experimental Philosophy of Mind

Many challenges to Gray, Gray, and Wegner's mental state attribution study have been raised over the past decade. The states don't seem well controlled. Use Malle (2019)

A feature of the early work: dogmatic

But, actually there is diversity in people's attributions of phenomenal states:

1. Individuals can be made to think in disembodied ways. (Salomons et al. 2021)
2. Any cultural diversity effects
3. System 1 and System 2 diversity

There is so far little agreement about what kinds of mental states people ascribe to robots. Gray, Gray and Wegner (2007) found that people tend to attribute the behavior of robots to mental states related to agency (e.g., memory, planning, and thought) but not subjective experience (e.g., fear, pain, and pleasure). Sytsma and Machery (2010) found, in contrast, that people refrain from attributing subjective states that have hedonic value for the subject, that is, valenced states (e.g., feeling pain and anger) as opposed to unvalenced states (e.g., smelling a banana or seeing red).

Buckwalter and Phelan (2013) further showed that people's tendency to attribute (or not) experiential or valenced states depends on the described function of the robot. Fiala et al. (2014) found that respondents in their experiments – when allowed to choose between different ways of describing the capabilities of a robot (e.g., the robot “identified the location of the box” vs. “knew the location of the box”) – preferred not to attribute mental states at all. The authors noted that responses to questions about the mental states of robots are influenced by a wide variety of factors, including the apparent function of the robot, the way in which the question is asked, and cultural platitudes about robots. In sum, it seems problematic to identify what kinds of mental states people ascribe to robots by asking them directly. Part of the problem, we believe, is that such questions are ambiguously open to interpretation as regarding the reality of the mental states of robots. As pointed out previously, people tend to predict and explain robot behavior with reference to mental states without reflecting on the reality of those states. Thus, when asked directly, a person might deny that a robot has a mind, despite having previously attributed mind to it upon being asked to describe its behavior (Fussell et al., 2008). (Thiellman and Ziemke)

Reflect on the large number of topics in analytic philosophy that involve discussions of when and why some variety of mental state is (or is properly) attributed to some entity. Such considerations clearly figure prominently in the mind-body problem and many other special topics in analytic philosophy of mind. However, they are also prominent in adjacent areas of philosophy of action, philosophy of language, ethics, and epistemology. What's more, philosophical discussions of mental state attribution vary in terms of their granularity. Some arguments focus specifically on the attribution of one particular kind of mental state—say, pain or intent—whereas others focus on the attribution of mental states as such or on some specific subset of mental states—say, propositional attitudes or phenomenally conscious mental states.

We see the same breadth and variety in the experimental philosophy of mental state attribution. There are experimental papers investigating the attribution of particular mental states (such as pain, intent, and happiness) and there are papers investigating the attribution of expansive kinds of mental states (such as phenomenal states, propositional attitudes, and perceptual states). And, whereas some theorists design studies with the primary aim of investigating how ordinary people attribute mental states, others empirically investigate mental state attribution for more instrumental reasons, perhaps to find out how people think about moral responsibility or freedom of the will. In this section, I will examine several disparate projects to construct a picture of the experimental philosophy of mental state attribution.

Were we to entertain a category of analytic philosophy called the philosophy of mental state attribution, we would quickly recognize just how many topics c

Think of the large variety of projects that arguably fall under the heading of the philosophy of mental state attribution.

(However, I will largely ignore the massive and still growing literature on attributions of intentionality, which deserves its own treatment.¹¹)

Peressini (2014) provided evidence for the claim that lay people tend to exclude robots from the category of experiencers (agents possessing phenomenal consciousness). In his study, participants were first presented with a description of what experiencers and non-experiencers are, after which they were asked to judge 20 entities on a 7-point scale (1 — clearly not an experiencer, 4—unsure, and 7—clearly an experiencer). Results indicate that robots, androids, and other artifacts are considered non-experiencers.

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Emotions and the body. Testing the subtraction argument

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